A Research Plan for the Development of a Graph-Informed Multimodal Attention Network (GIMAN) for Parkinson's Disease Prognosis

A New Paradigm for Parkinson's Prognosis: Integrating Population Graphs with Multimodal Deep Learning

Introduction and Rationale

Parkinson's disease (PD) is a neurodegenerative disorder characterized by profound clinical heterogeneity. Patients exhibit significant variability in symptom manifestation, rates of motor progression, and the onset and severity of non-motor symptoms, particularly cognitive decline.[1] This heterogeneity poses a substantial challenge for clinical management, patient counseling, and the design of therapeutic trials. A fundamental limitation in the development of effective, disease-modifying therapies is the inability to accurately predict individual patient trajectories and stratify cohorts into more homogeneous subgroups. The development of robust prognostic biomarkers is therefore a critical unmet need in PD research.

In recent years, machine learning (ML) and artificial intelligence (AI) have emerged as powerful tools for integrating the complex, high-dimensional data required to model PD progression. Early efforts demonstrated the potential of fusing multimodal data—such as neuroimaging, genetic profiles, and clinical information—to improve diagnostic accuracy over unimodal approaches.[2] These initial models often employed feature-level fusion, where features from different modalities are concatenated before being fed into a classifier, or decision-level fusion, where predictions from separate unimodal models are combined in an ensemble.[2] The advent of deep learning has further advanced this field, with architectures like Convolutional Neural Networks (CNNs) for imaging, Recurrent Neural Networks (RNNs) for longitudinal data, and Transformers for genetic data enabling the automatic extraction of complex, hierarchical features.[4] These models have achieved high accuracy in distinguishing PD patients from healthy controls and have begun to uncover latent relationships between

genotypes and clinical phenotypes.[4]
Despite this progress, a critical and unaddressed limitation pervades the current landscape of AI in neurology: existing models almost universally treat each patient as an independent and identically distributed (i.i.d.) data point. This assumption ignores the rich, latent relational structure within a patient cohort. Patients are not islands; they may share subtle similarities in their genetic risk profiles, molecular signatures, or patterns of neurodegeneration that place them into natural, data-driven subgroups. Ignoring these inter-patient relationships discards a powerful source of information that could regularize model training, improve generalization to unseen patients, and reveal novel disease subtypes.
This research plan is founded on a central hypothesis that directly confronts this limitation: **Modeling the latent structure of the patient cohort as a graph and leveraging this structure to inform the learning process for individual patients will lead to more robust, generalizable, and interpretable prognostic models for Parkinson's disease.** This approach represents a paradigm shift from purely patient-centric prediction to a population-informed framework. By constructing a patient similarity graph, we can explicitly model the relationships between individuals, allowing the model to learn not only from a single patient's data but also from the data of their most similar peers. This is analogous to a clinical expert's reasoning, where the prognosis for a new patient is informed by experience with previous, similar cases.

The GIMAN Framework: A Conceptual Overview

To operationalize this hypothesis, this plan proposes the development of a novel architecture: the **Graph-Informed Multimodal Attention Network (GIMAN)**. This framework is meticulously designed to synergistically combine the strengths of two cutting-edge AI methodologies. First, it employs Graph Neural Networks (GNNs), which are purpose-built to learn from graph-structured data, to model population-level information and inter-patient relationships.[6] Second, it utilizes a suite of powerful, modality-specific deep learning encoders, including 3D CNNs and Transformers, to extract salient, high-level features from the complex longitudinal data of each individual patient.[4]
The GIMAN framework is conceptualized as a three-stage pipeline that mirrors a sophisticated diagnostic and prognostic process:

1. **Patient Similarity Graph Construction:** The process begins by transforming the patient cohort from a simple collection of data points into a structured graph. Nodes in this graph represent individual patients, and weighted edges represent the degree of similarity between them, calculated from a curated set of baseline genetic, proteomic, and clinical data.
2. **Multimodal Feature Encoding:** In parallel, a series of specialized deep learning encoders process each patient's longitudinal, multimodal data. A hybrid CNN-RNN architecture will model the spatiotemporal evolution of neuroimaging data (sMRI and DAT-SPECT), a Transformer will capture complex epistatic interactions in genomic data,

and an RNN will map the trajectory of clinical scores.

3. **Graph-Attention Fusion for Prediction:** This is the core innovation of GIMAN. The rich feature representations from the encoders are integrated as node attributes in the patient graph. A Graph Attention Network (GAT) then propagates information across the graph, allowing each patient's representation to be refined by its most relevant neighbors. Finally, a cross-modal attention mechanism dynamically weighs the importance of each data stream before making a final prognostic prediction.

This hierarchical and integrative approach moves beyond simple data fusion. It first learns the trajectory of biomarkers within a patient, then situates that patient within a population of their peers, and finally synthesizes all available evidence to forecast their clinical future.

Specific Aims and Novelty

This research program is structured around four specific aims, each designed to build upon the last, culminating in a validated and interpretable prognostic tool.

- **Aim 1: To construct a heterogeneous patient similarity graph from the Parkinson's Progression Markers Initiative (PPMI) cohort using baseline genetic, proteomic, and clinical data to capture underlying patient subtypes and disease structures.** This foundational step transforms the dataset into a format amenable to graph-based learning, moving beyond the limiting i.i.d. assumption.
- **Aim 2: To develop and train the GIMAN architecture, a novel deep learning framework that integrates longitudinal multimodal data (neuroimaging, genetics, clinical scores) for the prognostic prediction of key clinical outcomes, namely motor progression (change in MDS-UPDRS Part III) and cognitive decline (conversion to Mild Cognitive Impairment or dementia).** This aim focuses on the core technical development of the model, integrating state-of-the-art components into a cohesive and powerful prognostic engine.
- **Aim 3: To rigorously validate GIMAN against a comprehensive suite of state-of-the-art unimodal and multimodal baselines using a nested cross-validation framework.** This ensures an unbiased and robust evaluation of the model's performance, directly addressing the high standards of methodological rigor required for top-tier publication and clinical translation.
- **Aim 4: To employ a multi-scale interpretability framework (GNNExplainer, SHAP, Grad-CAM) to deconstruct the GIMAN model's predictions, thereby identifying novel prognostic biomarkers, influential patient archetypes, and salient neurobiological features.** This aim is critical for translating the model's "black box" predictions into clinically actionable insights and generating new, data-driven hypotheses about PD pathophysiology.[4]

The **novelty** of this research plan is multi-faceted. While GNNs are an emerging area of interest in medical research [7], the GIMAN framework is, to our knowledge, the first to propose their application to a patient similarity graph for *prognostic* modeling in PD using *longitudinal, multimodal* data. It explicitly moves beyond the i.i.d. assumption that constrains virtually all current approaches. Furthermore, its hierarchical, gradual fusion architecture and multi-scale interpretability plan represent a significant advance in both technical sophistication and clinical relevance over existing models. The successful execution of this plan could redefine the state-of-the-art in computational neurology. Instead of merely asking "What will this patient's progression be?", GIMAN asks "Given this patient's unique biomarker trajectory, and their relationship to the broader landscape of Parkinson's disease phenotypes, what is their most likely future?". This population-informed perspective has profound implications. For instance, it could fundamentally reshape how patients are selected for clinical trials. Current stratification often relies on single biomarkers, such as GBA mutation status, or broad clinical criteria.[12] A model like GIMAN, however, naturally identifies data-driven patient communities within its graph structure. A clinical trial for a therapy targeting a specific inflammatory pathway could recruit patients from the specific graph community where proteomic markers of inflammation are identified as the most influential drivers of progression. This represents a truly multimodal, data-driven approach to trial enrichment, a significant leap towards the goal of precision medicine in Parkinson's disease.

Defining the Longitudinal PPMI Analysis Cohort

Data Source: The Parkinson's Progression Markers Initiative (PPMI)

The foundation of this research is the Parkinson's Progression Markers Initiative (PPMI) database, a landmark longitudinal, multicenter, observational study meticulously designed to identify and validate biomarkers of Parkinson's disease progression.[12] The unparalleled richness and scope of the PPMI dataset make it the ideal resource for developing and validating a sophisticated multimodal model like GIMAN. The study provides harmonized, high-quality data across a wide array of modalities, including:

- **Neuroimaging:** High-resolution 3T structural Magnetic Resonance Imaging (sMRI) and Dopamine Transporter Single-Photon Emission Computed Tomography (DAT-SPECT).
- **Genetics:** Genome-wide Single Nucleotide Polymorphism (SNP) data.
- **Biospecimens:** Cerebrospinal fluid (CSF) and plasma proteomics, as well as RNA sequencing data from whole blood.
- **Clinical Assessments:** Comprehensive and longitudinal evaluations of motor function (MDS-UPDRS), cognitive status (e.g., Montreal Cognitive Assessment - MoCA), non-motor symptoms (e.g., University of Pennsylvania Smell Identification Test - UPSIT; Scale for Outcomes in Parkinson's Disease-Autonomic - SCOPA-AUT), and behavioral changes.[12]

All data for this project will be formally requested and accessed through the LONI Image &

Data Archive (ida.loni.usc.edu), contingent upon the submission of a detailed online application and adherence to the PPMI Data Use Agreement and Publications Policy.[14] This ensures compliance with the ethical and data-sharing standards of the initiative.

Subject Selection and Cohort Definition

To ensure the integrity of the analysis and the clinical validity of the findings, a stringent, rule-based procedure will be implemented for cohort selection. This procedure is designed to maximize diagnostic certainty and data completeness, mirroring the high standards of the PPMI study itself.

Inclusion Criteria (PD Cohort):

- Participants must have a clinical diagnosis of PD established within two years of their baseline visit.
- Participants must be de novo, i.e., untreated with any dopaminergic medications at their baseline assessment.
- Crucially, all participants in the PD cohort must have imaging evidence of a dopaminergic deficit, as confirmed by their baseline DAT-SPECT scan. This is a critical quality control measure to rigorously exclude subjects with Scans Without Evidence of Dopaminergic Deficit (SWEDDs), whose underlying pathology may differ from typical PD.[12]
- Participants must have complete baseline data available for all core modalities required for the GIMAN framework: 3T T1-weighted sMRI, DAT-SPECT, and genome-wide SNP data.
- For longitudinal prognostic modeling, participants must have at least two post-baseline follow-up visits (e.g., at 12 and 24 months) with complete MDS-UPDRS Part III and MoCA scores.

Inclusion Criteria (Healthy Control Cohort):

- Participants must be free from any significant neurological dysfunction at baseline.
- Participants must have a baseline MoCA score greater than 26, indicating normal cognitive function.
- Participants must not have a first-degree relative with PD.
- Participants must have complete baseline data available for the core modalities (sMRI, DAT-SPECT, SNP).

Exclusion Criteria (All Subjects):

- Participants with significant medical or psychiatric comorbidities that could confound the analysis of PD progression will be excluded.
- Any imaging data flagged for poor quality during the central PPMI quality control process will be excluded.
- Subjects with a high percentage of missing data across key longitudinal clinical variables will be excluded to ensure the reliability of the prognostic endpoints.

Definition of Prognostic Endpoints

The GIMAN model will be trained and evaluated on two primary, clinically meaningful prognostic tasks that capture the core dimensions of PD progression.

1. **Motor Progression (Regression Task):** The model will be trained to predict the rate of motor decline. This will be operationalized as the slope of the MDS-UPDRS Part III (motor examination) score, calculated via linear regression for each patient over a 36-month period from their baseline visit. The UPDRS-III is the gold standard for assessing motor severity in PD, and its rate of change is a key outcome measure in clinical trials.[9] The output of this task will be a continuous variable representing the predicted annual change in UPDRS-III points.
2. **Cognitive Decline (Classification Task):** The model will be trained to predict the risk of significant cognitive decline. This will be framed as a binary classification task: predicting whether a patient will convert from a state of normal cognition to Mild Cognitive Impairment (MCI), or from MCI to PD-Dementia, within a 36-month follow-up window. Cognitive impairment is a major determinant of quality of life and disability in PD, making its prediction a high-priority clinical goal.[17]

Handling Missing Data

Longitudinal clinical studies are invariably affected by missing data due to participant attrition or missed assessments. Naive handling of missing data can introduce significant bias and reduce statistical power. To address this methodologically critical issue, a sophisticated imputation strategy will be employed. First, any feature with more than 50% missing values across the cohort will be excluded from the analysis.[16] For the remaining features with lower rates of missingness,
Multivariate Imputation by Chained Equations (MICE) will be used. Unlike simple mean, median, or mode imputation, MICE is an advanced technique that models each variable with missing values as a function of the other variables in the dataset. It performs a series of regression models and imputes the missing values iteratively. This approach is superior because it preserves the complex inter-variable relationships and correlations that are fundamental to a multimodal analysis, leading to more accurate and unbiased imputations.[16]
To provide a clear overview of the final cohort that will be used for model development, the following descriptive table will be generated.

Characteristic	Parkinson's Disease (PD) (N ≈ 400)	Healthy Controls (HC) (N ≈ 200)
Demographics		
Age (years, mean ± SD)	61.5 ± 9.8	60.9 ± 11.2
Sex (% Male)	65%	64%
Education (years, mean ± SD)	15.8 ± 3.1	16.2 ± 2.9

Baseline Clinical		
Disease Duration (months, mean ± SD)	7.2 ± 5.5	N/A
MDS-UPDRS Part III (mean ± SD)	22.8 ± 10.1	N/A
MoCA Score (mean ± SD)	28.1 ± 1.9	29.0 ± 1.1
Data Availability (N, %)		
Baseline (BL)		
sMRI	400 (100%)	200 (100%)
DAT-SPECT	400 (100%)	200 (100%)
SNP Genotyping	400 (100%)	200 (100%)
CSF Proteomics	380 (95%)	190 (95%)
24 Months (V06)		
sMRI	360 (90%)	185 (92.5%)
UPDRS-III / MoCA	370 (92.5%)	188 (94%)
36 Months (V08)		
UPDRS-III / MoCA	350 (87.5%)	180 (90%)
Table 1: Proposed structure for summarizing the demographics and longitudinal data availability of the final analysis cohort. Values are illustrative.		

This table serves a crucial function by transparently presenting the characteristics of the study population. For any peer reviewer, understanding the cohort is the first step in evaluating the research. This summary demonstrates a rigorous and well-defined selection process, justifies the choice of a 36-month prognostic window, and acknowledges the reality of data attrition, which the proposed methodological framework is designed to handle. This level of transparency is essential for establishing the credibility and potential generalizability of the study's findings.

Architectural Blueprint of the Graph-Informed Multimodal Attention Network (GIMAN)

Overview of the GIMAN Pipeline

The GIMAN architecture is a multi-stage, deep learning pipeline designed to integrate patient-level multimodal data with population-level relational information. The design philosophy is guided by the structured Multimodal Graph Learning (MGL) blueprint, which provides a formal framework for combining graph structures with other data types.[18] This framework consists of four conceptual components that map directly onto the GIMAN pipeline:

1. **Identifying Entities:** Each patient in the PPMI cohort is defined as a distinct entity or node.
2. **Uncovering Topology:** The latent relationships between patients are made explicit by constructing a patient similarity graph, defining the network topology.
3. **Propagating Information:** A GNN is used to pass messages along the edges of the graph, allowing the features of each patient to be informed by their neighbors.
4. **Mixing Representations:** The graph-refined, multimodal representations are fused using an attention mechanism to produce a final embedding for prognostic prediction.

This structured approach ensures that each component of the architecture has a clear and distinct purpose, contributing to a cohesive and powerful whole. A detailed flowchart will visually articulate the flow of data from raw inputs through each of these stages to the final output.

Stage I: Constructing a Heterogeneous Patient Similarity Graph

The initial and foundational stage of the GIMAN framework is the transformation of the tabular patient cohort into a graph-structured dataset. This is essential for enabling the use of GNNs and moving beyond the i.i.d. assumption.[6]

Objective:

The goal is to construct a weighted, undirected graph G=(V,E), where the set of vertices V corresponds to the set of patients in the cohort, and the set of edges E represents the similarity relationships between pairs of patients.

Methodology:

1. **Feature Selection for Similarity Definition:** To construct the graph, a specific subset of *baseline, non-imaging* features will be selected. This careful selection is crucial to avoid data leakage, ensuring that the information used to define the graph structure is distinct from the longitudinal imaging data that will be processed by the downstream deep learning encoders. The selected features will provide a holistic, multi-domain snapshot of each patient at their initial visit:
    - **Genetic Profile:** A Polygenic Risk Score (PRS) will be computed for each patient from their genome-wide SNP data. The PRS aggregates the effects of many common genetic variants into a single score that quantifies an individual's inherited risk for PD.
    - **Proteomic Profile:** A vector of key CSF biomarker concentrations will be used, including α-synuclein, amyloid-beta 42 (Aβ42), total-tau (t-tau), and

phosphorylated-tau (p-tau). These proteins are central to the pathophysiology of neurodegeneration and provide a molecular signature of the disease state.[14]

  - **Clinical Phenotype:** A vector of baseline non-motor clinical scores that capture key aspects of the disease phenotype will be included. This will feature the UPSIT score (olfactory function) and the SCOPA-AUT total score (autonomic dysfunction), both of which are known to be affected early in PD.[12] Demographic information such as age and sex will also be included.

2. **Similarity Metric Calculation:** For every pair of patients (i,j) in the cohort, a similarity score Sij will be computed. This will be a composite score, calculated as a weighted sum of similarities across the three domains: Sij=wgen·Simgen(i,j)+wprot·Simprot(i,j)+wclin·Simclin(i,j). The similarity within each domain (Sim) will be calculated using a radial basis function (RBF) kernel on the standardized feature vectors, which is effective at capturing non-linear relationships. The weights (w) can be either pre-defined based on domain expertise or learned as part of the model's training process.
3. **Graph Sparsification:** A fully connected graph, where every patient is linked to every other patient, is computationally expensive and informationally noisy. To create a more meaningful and tractable graph, a k-nearest neighbor (k-NN) approach will be used for sparsification.[6] For each patient node, an edge will only be created to its top *k* most similar neighbors (based on the score Sij). The weight of the edge will be the similarity score itself. This results in a sparse graph that captures the most significant local relationships within the patient cohort.

Stage II: Modality-Specific Longitudinal Feature Encoding

This stage runs in parallel to the graph construction and involves a set of sophisticated deep learning encoders, each tailored to extract rich, high-level feature representations from the unique structure of a specific data modality. This architecture embodies the principles of intermediate fusion, where features are first learned within each modality before being combined.[22]

Spatiotemporal Imaging Encoder (for sMRI & DAT-SPECT)

- **Architecture:** A hybrid 3D Convolutional Neural Network (CNN) followed by a Gated Recurrent Unit (GRU).
- **Rationale:** This two-part architecture is designed to capture both the spatial and temporal dimensions of the neuroimaging data. The 3D CNN component, likely a lightweight variant of a Residual Network (ResNet), is adept at processing volumetric data. It will be applied to each 3D image (sMRI and DAT-SPECT) at each available time point to extract a compact feature vector that represents the key spatial patterns of

brain structure and dopaminergic function at that moment.[4] The sequence of these spatial feature vectors over time (e.g., Baseline, Year 1, Year 2) forms a trajectory of neurodegeneration. This sequence is then fed into a GRU, a type of recurrent neural network chosen for its computational efficiency and effectiveness in modeling temporal dependencies and learning the dynamics of disease progression.[25] To handle the two imaging modalities, separate initial convolutional layers will process the sMRI and DAT-SPECT data independently, after which their feature maps will be concatenated and passed through a shared set of deeper CNN and GRU layers.

- **Preprocessing:** Rigorous and standardized preprocessing is paramount for reliable imaging analysis. sMRI data will undergo a standard pipeline including skull stripping, bias field correction, intensity normalization, and non-linear registration to a common template space (e.g., MNI152). DAT-SPECT data will follow a well-established pipeline consistent with PPMI protocols, involving spatial normalization to a template and intensity normalization (e.g., to a reference region like the cerebellum or occipital cortex) to convert raw counts into semi-quantitative striatal binding ratios (SBRs).[16] This standardization is critical for pooling data from multiple imaging centers.

Genomic Transformer Encoder (for SNP data)

- **Architecture:** A Transformer-based model, conceptually similar to the GWAS-Transformer architecture.[4]
- **Rationale:** The primary challenge in analyzing genomic data is modeling the complex, non-linear interactions (epistasis) among thousands or millions of genetic variants. Traditional ML methods often struggle with this high dimensionality and fail to capture these subtle interactions. The self-attention mechanism at the core of the Transformer architecture is uniquely suited for this task. It can learn long-range dependencies and interactions between SNPs across the entire genome without being constrained by physical proximity on a chromosome, thereby addressing a major limitation in conventional genetic association studies.[4]
- **Input:** To manage the high dimensionality of SNP data, a feature selection step will be applied. SNPs passing a liberal p-value threshold (e.g., $p<10{-}4$) from a preliminary Genome-Wide Association Study (GWAS) will be selected. This reduces the input space while retaining potentially relevant signals. These selected SNPs will be converted into learned embeddings and fed as a sequence into the Transformer encoder.

Clinical Trajectory Encoder (for scalar clinical data)

- **Architecture:** A separate, smaller GRU network.
- **Rationale:** To explicitly model the progression of the clinical phenotype, a dedicated GRU will process the multivariate time series of scalar clinical assessments (e.g.,

MDS-UPDRS Parts I & II, MoCA, UPSIT, etc.). This encoder will learn the temporal dynamics of symptom evolution, creating a feature representation that captures the patient's clinical trajectory in parallel with their biological (imaging and genetic) data.

Stage III: Graph-Attention Fusion and Prognostic Prediction

This final stage is the innovative core of the GIMAN framework, where the population-level graph structure is integrated with the patient-level multimodal features to generate a final, informed prediction.

- **Architecture:** A Graph Attention Network (GAT) followed by a cross-modal attention fusion module.
- **Rationale and Mechanism:** The fusion process is a carefully orchestrated, multi-step sequence:
    1. **Node Feature Initialization:** The powerful, high-level feature vectors generated by the three parallel encoders in Stage II are concatenated for each patient. This concatenated vector becomes the initial feature representation for the corresponding node in the patient similarity graph created in Stage I.
    2. **Graph Information Propagation:** A GAT layer is then applied to this feature-rich graph. Unlike simpler Graph Convolutional Networks (GCNs) that aggregate information from neighbors uniformly, a GAT uses a self-attention mechanism to learn the relative importance of each neighbor.[21] It computes attention coefficients for each connected node, allowing it to dynamically weight and aggregate information from the most influential peers in the graph. This step explicitly leverages the population structure to refine and regularize each patient's learned embedding, effectively "smoothing" representations based on community structure and propagating prognostic signals across similar patients.[18]
    3. **Cross-Modal Attention Fusion:** After the GAT layer has produced a graph-informed embedding for each patient, this embedding still contains distinct segments corresponding to the imaging, genomic, and clinical modalities. These segments are fed into a final cross-modal attention fusion module. Inspired by architectures like the "Mutual Attention" block [31] and other cross-attention designs [32], this module learns to dynamically assign importance weights to each of the feature streams. For example, for a patient in a graph community characterized by high genetic risk, the model might learn to up-weight the contribution of the genomic encoder's features. Conversely, for a patient whose neighbors show rapid motor decline, the features from the spatiotemporal imaging encoder might be given higher importance.
    4. **Prediction Head:** The final, context-aware, and dynamically weighted feature vector is passed to a small multi-layer perceptron (MLP) that serves as the prediction head. The output layer of the MLP is tailored to the specific prognostic task: a single neuron with a linear activation function for the motor progression

regression task, and a single neuron with a sigmoid activation function for the binary cognitive decline classification task.

This architectural design represents a sophisticated form of "gradual fusion," a concept identified as a promising but underexplored frontier in multimodal deep learning.[22] Information is fused at multiple, increasingly abstract levels: first *within* modalities over time (via the GRUs), then *across* similar patients in the cohort (via the GAT), and finally *across* the different modalities themselves (via the attention fusion module). This hierarchical approach is more powerful and potentially more biologically plausible than simple, one-shot fusion methods like feature concatenation. Furthermore, the modularity of the design offers a significant practical advantage. As the PPMI study evolves and new data modalities become available (e.g., digital biomarkers from wearables [34] or advanced proteomics), new modality-specific encoders can be developed and seamlessly "plugged into" the GIMAN framework. This ensures the architecture remains adaptable and future-proof, capable of incorporating new streams of data without requiring a complete redesign of the core fusion logic.

A Framework for Rigorous, Unbiased Model Validation

A novel architecture is only as valuable as the rigor with which it is evaluated. To preemptively address the stringent methodological critiques of top-tier journals, this research plan incorporates a comprehensive and robust validation framework designed to produce unbiased performance estimates and demonstrate clear superiority over existing methods.

Nested Cross-Validation for Unbiased Performance Estimation

A common pitfall in machine learning research is "information leakage," where information from the test set inadvertently influences the model training process, typically during hyperparameter tuning. This leads to overly optimistic performance estimates that do not generalize to new, unseen data. To completely avoid this issue, a **nested 10-fold cross-validation (CV)** strategy will be employed.[2]

- **Outer Loop (Performance Estimation):** The entire dataset will be partitioned into 10 equally sized folds. The outer loop will iterate 10 times. In each iteration, one fold is held out as the final, untouched test set, while the remaining 9 folds are passed to the inner loop for model development. The performance of the final model is calculated on this held-out test fold. The ultimate performance metrics for the GIMAN framework will be the average and standard deviation of the scores obtained across these 10 test folds.
- **Inner Loop (Hyperparameter Tuning):** Within the 9 folds used for training in each outer loop iteration, a separate, inner cross-validation process (e.g., a 5-fold CV) will be conducted. This inner loop is used exclusively for hyperparameter optimization. A grid search or Bayesian optimization will be performed to find the best set of

hyperparameters (e.g., learning rate, number of attention heads in the GAT, dropout rates, etc.) by evaluating performance on the inner validation sets. Once the optimal hyperparameters are identified, a new model is trained from scratch on all 9 training folds using these parameters.

- **Justification:** This nested structure ensures a strict separation between the data used for hyperparameter selection and the data used for final performance evaluation. The 10-fold split for the outer loop is a widely accepted standard that provides a good balance between the bias and variance of the performance estimate for datasets of the size anticipated from PPMI.[35] This rigorous methodology is a hallmark of high-quality machine learning research and is essential for producing credible and reproducible results.

Addressing Class Imbalance in Cognitive Decline Prediction

The classification task of predicting cognitive decline is expected to feature a significant class imbalance, with far more patients remaining cognitively stable ("non-converters") than those who decline ("converters"). In such scenarios, a model that simply maximizes accuracy will perform poorly, as it can achieve a high score by always predicting the majority class.[38] To address this, a multi-pronged strategy focused on both the loss function and data sampling will be adopted.

- **Primary Strategy (Focal Loss):** The primary approach will be the use of a **focal loss function** during model training. Focal loss is an enhancement of the standard cross-entropy loss that dynamically down-weights the contribution of well-classified examples (typically the majority class) to the total loss. This forces the model to focus its learning on the more challenging, hard-to-classify examples, which are often the minority class instances.[39]
- **Comparative Analysis and Ablation Studies:** To demonstrate a thorough investigation of the problem, the performance of focal loss will be compared against several alternative techniques in an ablation study:
  - **Weighted Cross-Entropy Loss:** A simpler approach where a higher static weight is assigned to the minority class in the loss calculation, effectively increasing the penalty for misclassifying it.[38]
  - **Advanced Synthetic Oversampling (ADASYN):** This technique is an improvement over the standard SMOTE algorithm. ADASYN generates more synthetic samples for minority class instances that are located near the decision boundary and are thus harder to learn, adaptively shifting the classification boundary to focus on difficult examples.[40]
  - **Random Undersampling:** While less desirable due to the potential loss of valuable information, randomly removing samples from the majority class will be included as a baseline balancing technique.[38]

Comprehensive Performance Metrics

The evaluation of the model's performance will not rely on a single metric. Instead, a comprehensive suite of metrics will be used to provide a nuanced and holistic assessment of its capabilities for both prognostic tasks.

- **For Classification (Cognitive Decline):**
  - **Area Under the Receiver Operating Characteristic Curve (AUC-ROC):** A measure of the model's overall discriminative ability.
  - **Area Under the Precision-Recall Curve (AUC-PR):** A more informative metric than AUC-ROC for imbalanced datasets, as it evaluates the trade-off between precision and recall.
  - **F1-Score:** The harmonic mean of precision and recall, providing a single score that balances both concerns.
  - **Precision (Positive Predictive Value):** The proportion of positive predictions that are truly positive.
  - **Recall (Sensitivity):** The proportion of actual positive cases that are correctly identified.
  - **Specificity:** The proportion of actual negative cases that are correctly identified.[2]
- **For Regression (Motor Progression):**
  - **Coefficient of Determination (R2):** The proportion of the variance in the outcome that is predictable from the features.
  - **Mean Absolute Error (MAE):** The average absolute difference between the predicted and actual values, providing an easily interpretable measure of error in the original units of the UPDRS score.
  - **Root Mean Squared Error (RMSE):** The square root of the average of squared differences, which penalizes larger errors more heavily.

Benchmarking Against State-of-the-Art

To unequivocally demonstrate the novelty and superior performance of the GIMAN framework, it will be rigorously benchmarked against a series of progressively more complex and relevant baseline models. This comparative analysis is essential for contextualizing the results and substantiating claims of a scientific advance.

- **Unimodal Baselines:** Each of the modality-specific encoders (Spatiotemporal Imaging, Genomic Transformer, Clinical Trajectory) will be trained and evaluated as standalone models. This will establish a performance baseline for each individual modality and quantify the benefit of multimodal fusion.
- **"Shallow" Machine Learning Baseline:** A powerful, non-deep-learning model, specifically an **XGBoost** classifier/regressor, will be trained. This model will use a carefully engineered, "flattened" feature vector comprising curated variables from each

modality (e.g., striatal binding ratios from DAT-SPECT, hippocampal volumes from sMRI, the polygenic risk score, and key clinical variables). This provides a strong baseline representing current best practices in classical machine learning.[42]

- **Simple Deep Learning Fusion Baseline:** A deep learning model that lacks the novel graph and attention components of GIMAN. This model will concatenate the feature vectors produced by the three modality-specific encoders and feed them directly into a standard MLP for prediction. This will isolate and quantify the performance gain attributable specifically to the graph-based and attention-based fusion mechanisms.
- **Published State-of-the-Art (SOTA) Baselines:** To provide the most direct and compelling comparison, key high-performing models from recent literature that also utilized the PPMI dataset will be re-implemented. This will include models such as the **AE_Stacking** framework, which employed feature- and decision-level fusion of MRI and SNP data [2], and the
  RACF network, an interpretable deep learning framework that fused GWAS-prescreened SNPs with sMRI features.[4]

The results of this comprehensive benchmarking exercise will be summarized in a clear and concise table, allowing for a direct comparison of GIMAN's performance against all baselines.

Model	Motor Progression (Regression)	Cognitive Decline (Classification)
	MAE (UPDRS-III pts)	**R2**
Unimodal Baselines		
Imaging Encoder Only	3.5 ± 0.4	0.35 ± 0.05
Genomic Encoder Only	4.2 ± 0.5	0.20 ± 0.06
Clinical Encoder Only	3.8 ± 0.3	0.30 ± 0.04
Shallow ML Baseline		
XGBoost	3.2 ± 0.4	0.45 ± 0.06
DL Baselines		
Simple Fusion (Concat+MLP)	3.0 ± 0.3	0.50 ± 0.05
AE_Stacking (re-impl.)	N/A	N/A
RACF (re-impl.)	N/A	N/A
Proposed Model		
GIMAN	**2.5 ± 0.3**	**0.65 ± 0.04**
Table 2: Proposed structure for presenting the comparative performance of GIMAN against baseline models. All metrics reported as mean ± standard deviation across 10 outer cross-validation folds. Values are illustrative and represent hypothetical superior performance for GIMAN.		

This table is designed to be the central pillar of the results section. It provides the quantitative evidence needed to support claims of superiority, demonstrating not only that multimodal fusion is beneficial (by comparing to unimodal baselines) but also that the specific architectural innovations of GIMAN provide a significant performance advantage over both simpler fusion strategies and previously published state-of-the-art models.

Unveiling the "Black Box": Multiscale Interpretability for Biomarker Discovery

A highly accurate predictive model that operates as an unexplainable "black box" has limited value in a clinical and scientific context. For a model to be trusted by clinicians, accepted by regulatory bodies, and used as a tool for scientific discovery, its decision-making process must be transparent. Therefore, a core component of this research plan is a multi-scale interpretability framework designed to deconstruct the GIMAN model and translate its internal workings into meaningful biological and clinical insights.[4]

Rationale for a Multi-Scale Approach

The hierarchical and multimodal nature of the GIMAN architecture necessitates a multi-faceted interpretation strategy. No single interpretability method can fully explain its behavior. Therefore, the model will be probed at three distinct levels of analysis to provide a comprehensive understanding:

1. **Population Level:** To understand the learned structure of the patient cohort and identify key patient archetypes.
2. **Individual Patient Level:** To explain the specific factors driving the prognostic prediction for any given patient.
3. **Biomarker Level:** To pinpoint the most salient biological features within each data modality that the model has learned to associate with disease progression.

Global Interpretation: Identifying Patient Archetypes and Key Similarities

- **Method: GNNExplainer** or a similar graph attribution technique will be applied to the trained GAT layer of the GIMAN model.[11]
- **Objective:** GNNExplainer is an algorithm designed to identify the most influential components of a graph for a given prediction. By applying it globally, it can reveal the

macro-level disease landscape as learned by the model. This analysis can answer critical questions such as:

- **Which patient subgraphs are most important?** The analysis can highlight specific communities of patients within the graph (e.g., a cluster of patients with a specific genetic profile and rapid cognitive decline) that are particularly influential for the model's overall predictive accuracy. This provides a data-driven method for discovering and characterizing novel patient subtypes.
- **Which inter-patient similarities are most critical?** The method can identify the specific edges in the graph that are most crucial for information propagation. This helps to understand the basis of the learned patient subtypes—for example, revealing that similarity in CSF proteomic profiles is more important for defining certain prognostic groups than similarity in clinical symptoms.

Local Interpretation: Explaining Individual Patient Predictions

- **Method: SHapley Additive exPlanations (SHAP)** will be applied to the final output of the full GIMAN model.[4]
- **Objective:** SHAP is a powerful, game theory-based method that can compute the contribution of each input feature to an individual prediction. This provides granular, patient-specific explanations, which are essential for clinical translation.
  - **Personalized Biomarker Ranking:** For any given patient, a SHAP analysis can generate a "force plot" that visualizes which features are pushing their predicted risk up or down. This could reveal, for example, that a patient's high predicted risk of motor progression is driven primarily by an accelerated rate of dopamine loss in the putamen and a high polygenic risk score, despite their relatively mild baseline symptoms.
  - **Clinical Decision Support:** These individualized explanations form the basis of a trustworthy clinical decision support tool. A report could be generated for a clinician that not only provides a risk score but also explains the key factors contributing to that score, grounding the AI's prediction in concrete, understandable clinical and biological data.
  - **Global Feature Importance:** By averaging the absolute SHAP values for each feature across all patients in the test set, a global ranking of feature importance can be generated, identifying the most powerful prognostic markers overall.

Modality-Specific Interpretation: Pinpointing Salient Neurobiological Features

- **Method:** For the spatiotemporal imaging encoder, **Gradient-weighted Class Activation Mapping (Grad-CAM)** will be employed.[4]

- **Objective:** Grad-CAM produces a "saliency map" or "heatmap" that can be overlaid on an input image, highlighting the regions that the CNN component focused on most intensely when making its prediction.
  - **Biological Plausibility and Discovery:** This visual analysis can be used to validate that the model is learning biologically plausible patterns. For PD, one would expect the model to focus on canonical regions like the substantia nigra, putamen, and caudate nucleus. Grad-CAM can confirm this. More excitingly, it could also reveal novel, unexpected brain regions that are prognostically important, generating new hypotheses for neurobiological research. For instance, it might highlight specific cortical or cerebellar regions whose atrophy rates are highly predictive of cognitive decline.

The key findings from the interpretability analysis, particularly the global feature importance rankings from SHAP, will be synthesized into a table to clearly present the novel biomarkers identified by the model.

Modality	Feature	Mean Absolute SHAP Value (Contribution to Prediction)
Imaging	1. Rate of Change in Putamen DAT Binding	0.45
	2. Baseline Hippocampal Volume (ICV-corrected)	0.38
	3. Rate of Change in Caudate DAT Binding	0.31
Genomic	4. Polygenic Risk Score (PRS)	0.41
	5. SNP rs356181 (SNCA locus)	0.29
	6. GBA N370S carrier status	0.25
Proteomic (CSF)	7. Baseline α-synuclein concentration	0.35
	8. Baseline p-tau / Aβ42 ratio	0.32
Clinical	9. Baseline UPSIT Score	0.39
	10. Rate of Change in SCOPA-AUT Score	0.27
Table 3: Proposed structure for presenting the top-ranked predictive features identified by SHAP analysis, organized by data modality. Values are illustrative.		

This table is arguably as important as the performance table. For a high-impact scientific journal, the generation of new knowledge is paramount. This table distills the model's learned intelligence into a ranked list of data-driven, prognostically significant biomarkers. It bridges the gap between data science and clinical neurology, providing tangible, testable hypotheses for future research and demonstrating that the GIMAN model is not just a predictor, but a powerful engine for scientific discovery.

From Bench to Bedside: Pathways to Clinical Translation and Future Work

Synthesis and Discussion

This research plan outlines the development of GIMAN, a novel deep learning framework poised to significantly advance the prediction of disease progression in Parkinson's disease. By synergistically integrating population-level graph structures with patient-level longitudinal multimodal data, GIMAN is designed to overcome the fundamental limitations of current models that treat patients in isolation. The expected outcomes of this project are fourfold: (1) a highly accurate prognostic model for key clinical endpoints in PD; (2) a data-driven method for identifying and characterizing novel PD subtypes based on graph community structures; (3) a ranked list of multimodal biomarkers that are most predictive of disease progression; and (4) a robust, open-source framework that is extensible to new data modalities and other neurodegenerative diseases.

The potential clinical impact of this work is substantial. An accurate prognostic tool could revolutionize patient counseling, allowing clinicians to provide more personalized information about expected disease course. It could enable proactive management strategies, targeting interventions for patients identified as being at high risk for rapid motor or cognitive decline. Most significantly, GIMAN could serve as a powerful tool for enriching clinical trials. By stratifying patients based on their predicted trajectories or their membership in specific graph-defined communities, researchers could design smaller, shorter, and more statistically powerful trials, accelerating the development of truly disease-modifying therapies.

Limitations and Future Directions

A proactive acknowledgment of potential limitations is essential for credible scientific research. The primary limitation of this proposed study is its retrospective nature, relying on the existing PPMI dataset. While PPMI is a gold-standard cohort, it may contain inherent selection biases, and findings may not be perfectly generalizable to the broader PD population. The computational complexity of the GIMAN model is also a consideration, which may pose challenges for training and deployment.

These limitations directly inform the critical next steps and future directions for this research program:

- **External Validation:** The single most important future direction is the validation of the trained GIMAN model on a completely independent, external cohort. A suitable candidate would be the Parkinson's Disease Biomarkers Program (PDBP) cohort, which contains similar multimodal data.[43] Successful validation on an external dataset is the sine qua non for demonstrating the model's generalizability and robustness.
- **Exploration of Causal Inference:** The current framework is designed to identify predictive correlations. Future work could employ techniques from causal inference to begin disentangling these correlations and explore potential causal relationships between the identified biomarkers and the mechanisms of disease progression.
- **Development of a Clinical Decision Support Tool:** The ultimate goal is clinical translation. A future project would focus on distilling the complex GIMAN model into a simplified, computationally efficient version that could be deployed in a clinical setting. This would involve creating an intuitive user interface that provides not only a risk prediction but also the patient-specific SHAP-based explanations to support clinical decision-making.
- **Application to Other Disease States:** The GIMAN framework is conceptually agnostic to Parkinson's disease. Its architecture for integrating population graphs with longitudinal multimodal data could be readily adapted to study other complex, heterogeneous neurological disorders such as Alzheimer's disease, Amyotrophic Lateral Sclerosis (ALS), or Multiple Sclerosis (MS).

Conclusion

The research plan detailed herein presents a novel, rigorous, and clinically relevant strategy to leverage the full power of modern artificial intelligence for understanding and predicting Parkinson's disease. By moving beyond the patient-as-an-island paradigm and embracing the interconnectedness of the patient cohort, the GIMAN framework is designed to deliver a new level of prognostic accuracy and scientific insight. Through its sophisticated architecture, robust validation plan, and deep commitment to interpretability, this project has the potential to yield not only a state-of-the-art predictive tool but also new, data-driven knowledge about the fundamental nature of Parkinson's disease, paving the way for a future of more precise and personalized neurological care.

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